Do Now · retrieval
Sequence & branching in C#

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why would Unicode be chosen instead of standard ASCII for a multilingual application?

Q2

Convert 01011000 0011 to denary using an 8-bit two’s-complement mantissa and a 4-bit two’s-complement exponent.

Q3

Convert $6D_{16}$ to eight-bit binary.

Q4

Convert 11110110 from 8-bit two’s-complement binary to denary. Show signed place values.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Sequence = instructions in order — Sequence is the normal top-to-bottom execution of statements. Each statement happens after the previous one unless another construct changes the flow. — CODE:

```
Console.Write("Name: ");
string name = Console.ReadLine();
Console.WriteLine("Hello " + name);
```

Core idea 2

Input arrives as text — Console.ReadLine() returns a string. If the program needs a whole number, parse the input into an int . — CODE:

```
Console.Write("Age: ");
int age = int.Parse(Console.ReadLine());
```

Core idea 3

Branching chooses a route — An if condition must evaluate to true or false . Only the matching branch is executed. — CODE:

```
if (age >= 18)
{
Console.WriteLine("Adult");
}
else
{
Console.WriteLine("Under 18");
}
```

New Learning · applying the ideas

Core idea 4

Conditions and order matter — Use == to compare equality, not = . Use && , || and ! for AND, OR and NOT. — With else if , conditions are tested from top to bottom. Put the most demanding threshold first.

Vocabulary · use the correct technical terms

Sequence — Instructions executed in their stated order.

Branching / selection — Choosing a route through a program according to a condition.

Boolean expression — An expression that evaluates to true or false.

Condition — The test inside an if statement.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Build a result checker

Follow the worked method

1. Use the standard .NET Fiddle structure — CODE:
using System;

public class Program
{
public static void Main()
{
// code goes here
}
} — Main program instructions belong inside Main() .

2. Test the highest threshold first — CODE:
if (score >= 70)
{
Console.WriteLine("Distinction");
}
else if (score >= 50)
{
Console.WriteLine("Pass");
}
else
{
Console.WriteLine("Not yet");
} — A score of 85 satisfies both >= 70 and >= 50 , so the higher threshold must be tested first.

3. Use values that expose mistakes — CODE:
85 → Distinction
70 → Distinction
50 → Pass
49 → Not yet

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Build the missing decision

CODE:

```
Console.Write("Number: ");
int number = int.Parse(Console.ReadLine());

// ?
```

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Positive, negative or zero
Ask the user for an integer. Output exactly one of:
Positive · Negative · Zero

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Simple login
Ask for a username and a PIN.
Output Access granted only when the username is "admin" and the PIN is 2468 .
Otherwise output Access denied .

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In C#, == is used to compare two values for equality.

Use key vocabulary from memory

1. Explain Sequence in your own words.

2. Explain Branching / selection in your own words.

3. Explain Boolean expression in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.